

CAD v7.5 - defect repair pass

Date: 2026-07-30 **Trigger:** the v7.3 pre-assembly audit found 8 blocking defects. None had been fixed. **Scope:** everything fixable without calipers. Four items remain blocked on measurement and are now tagged in-source with MEASURE (Mnn) pointing at docs/MEASUREMENTS_NEEDED.md.

Every part was re-rendered and checked for Volumes: 2 - meaning one connected solid. That check is what caught the floating cable hooks in v7.4 and the vanished nav caps before that; Simple: yes alone does not catch disconnected geometry.

Side-by-side: what changed

#	File	Was	Now	Why
1	mech_layout.scad	homing magnet declared twice , in two files, already drifted (cam r=17.35 vs plate y=20)	one shared block: homing_mag_dia/thk/fit/r/angle	The sensor must sit under the magnet. Two copies guarantee they drift apart.
2	braille_cam.scad	magnet_depth = 2.0 into a 2.0mm floor	1.2 (3x1mm magnet), + assert	Pocket was a through-hole cratering tracks 2, 3 and 4. A foot crossing it jams.
3	braille_cam.scad	comment claimed "8mm engagement"	corrected to 6mm, full z-stack failure documented	Stale comment hid a real showstopper.
4	base_plate.scad	base_length = 56	58	Left ear screw had a 0.35mm wall - breaks out. Also fixed standoffs overhanging by 1mm.
5	base_plate.scad	spring_cavity() - 22x16 through-window	deleted	Right motor screw (x=+9.5) opened into it -> one-screw motor . Cavity was v6.x legacy; springs moved to the top plate in v7.1.
6	base_plate.scad	dia 4.3 clearance holes	dia 3.3 thread-forming pilots	A nut at x=+9.5 sits under the spinning cam. Screw now threads straight into the plate from below.
7	base_plate.scad	hall pocket cut down from the top at (0,20)	cut up from the underside at (0, 17.35)	Old pocket lay entirely inside the cam pocket at the same depth - it removed nothing; the feature did not exist on the printed part.
8	base_plate.scad	-	assert on sensor thickness	Fails loudly if the measured sensor won't fit, instead of printing a useless pocket.
9	esp32_pod_params.scad	pod_length = 64	68	Board overran the cavity by 1.75mm - would not go in. Bonus: pod is now a 68x68x58 cube matching a cell exactly.
10	esp32_pod_params.scad	devkit_x_offset = 4	-2	Plug reach into the pod cut from 6.25mm to 2.25mm.
11	esp32_pod_params.scad	usb_z = floor + strip_h + 1	board_under_z - 1.0	Arithmetic bug - ignored hdr_channel_depth. Hole sat ~2mm high; only 0.5mm overlapped the port.
12	esp32_pod_params.scad	USB cutout 10x7	13x9	A real cable's moulded body doesn't pass 10x7.

#	File	Was	Now	Why
13	esp32_pod_params.scad	barrel_jack_x = -22	-20	Cradle overran the wall; lid couldn't close.
14	esp32_pod_params.scad	lid_screw_x = 27.5 hard-coded	pod_int_length/2 - 0.5	See below - this one was created by fix #9.
15	esp32_pod_lid.scad	4 hard-coded dims matching the old pod	derived from pod_length/pod_width	Lengthening the pod would have silently desynced the lid.
16	esp32_pod_lid.scad	-	assert on cradle vs wall	The overrun was invisible because all three jack dims are invented.
17	outer_box.scad	muscle-board bosses always on	use_muscle_board_bosses = false	dia 4x4mm bosses sit inside the ULN2003's footprint - board would perch 4mm up in a 16mm pocket.
18	outer_box.scad, mid_plate.scad	motor numbers unmarked	tagged MEASURE (M1), (M2)	So the assumptions are findable rather than looking like facts.

The defect that fix #9 *created*

Worth recording, because it's the same bug the codebase has now hit three times.

lid_screw_x = 27.5 was correct **only** while pod_int_length was 56: the inner wall sat at x=28, so a dia 5 boss at 27.5 spanned 25-30 and embedded 2mm into the wall.

Lengthening the pod moved that wall to x=30. The boss would then have spanned 27-32 and merely **touched** the wall at exactly one plane - a disconnected body in the STL, printing as a loose tower. That is precisely the failure the comment block above that line was written to describe after it happened in v6.1b.

It is now a formula tied to the wall, so it cannot drift again.

Pattern: hard-coded numbers that "happen to match" another dimension are the main source of defects in this project. Fixes #1, #14 and #15 are all the same cure.

Verification

All parts re-rendered from source. Volumes: 2 = one connected solid:

Part	Simple	Volumes
outer_box	yes	2
mid_plate	yes	2
base_plate	yes	2
top_plate	yes	2
dot_insert	yes	2
linkage	yes	2
nav_cap	yes	2
pogo_end_cap	yes	2
esp32_pod_shell	yes	2
esp32_pod_lid	yes	2
braille_cam	yes	2

Both new asserts were exercised - the cam and lid render clean, so the magnet pocket leaves enough floor and the cradle fits inside the wall.

Still open - blocked on measurement

These are **not** fixable by guessing harder. Each depends on a physical number.

Defect	Needs	Consequence if ignored
Cam z-stack doesn't close: hub_h=4 into a 2mm gap	M2, M6, M7	All six dots permanently ~2mm proud - every dot reads as "on"
Hall pocket depth is 1.6mm	M11b	Sensor may not fit; assert will catch it
Barrel jack cradle is 100% invented	M12-M16	Jack unsupported, or lid won't close
ESP32 pin-row pitch: CAD says 38-pin, docs say 30-pin	M18, M21	ESP32 cannot plug into the pod at all

The z-stack must be re-derived in **one pass** once M2/M6/M7 arrive - motor height -> base plate -> cam -> linkage length are a single chain, and patching them individually just relocates the error.

Bill-of-materials correction

ELECTRONICS_BOM.md **item 7 says buy a 3x2mm homing magnet. Buy 3x1mm.**

The cam disc floor is 2.0mm. A 2mm magnet needs a 2mm pocket, which removes the entire floor. 1mm leaves 0.8mm behind and couples just as well, because the magnet still sits flush with the disc underside.

What did NOT change, deliberately

- **68x68mm cell shell, docking windows, magnet pockets, 4mm walls** - already physically fit-tested. Untouched.

- **Teardrop magnet pockets** - correct shape-based self-support. Keep.
- **No sacrificial bridges re-added** - removed in v7.4 for good reasons documented in `outer_box.scad`.

- **Nothing in the resin plate-1 batch already ordered.** All of v7.5's changes are in PETG parts not yet printed. **Nothing in transit is affected.**